SCHOOL OF ENGINEERING AND TECHNOLOGY

ASSIGNMENT Can you please provide me with access to this document? PROJECT SUBMISSION FORM

PROGRAMME: BSc (Hons) Information Systems (Data Analytics)

SEMESTER: Jan /close Apr / Aug 2024

SUBJECT: STAT1114/STAT2044 Statistics for Data Analytics

DEADLINE: 29 July 2024, 11.59 pm

INSTRUCTIONS TO CANDIDATES

- This is a group project.

IMPORTANT
The University requires students to adhere to submission deadlines for any form of assessment. Penalties are applied in relation to unauthorized late submission of work.

- **Coursework submitted after the deadline but within 1 week will be accepted for a maximum mark of 40%.**
- **Work handed in following the extension of 1 week after the original deadline will be regarded as a non-submission and marked zero.**

Lecturer's Remark (Use additional sheet if required)
List down the name of the group members and the student IDs here.

I Liang Yrenn Da (22067128) received the assignment and read the comments.

Liang Yrenn Da (13/7/2024)

I Howell Yek Teck Kee (22055479) received the assignment and read the comments.

Howell Yek Teck Kee (13/7/2024)

I Ooi Jin Yon (22039069) received the assignment and read the comments.

Ooi Jin Yon (13/7/2024)

I Yeoh Jie Long (22071112) received the assignment and read the comments.

Yeoh Jie Long (13/7/2024)

Academic Honesty Acknowledgement

"I *Liang Yrenn Da* verify that this paper contains entirely my own work. I have not consulted with any outside person or materials other than what was specified (an interviewee, for example) in the assignment or the syllabus requirements. Further, I have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. I realize the penalties *(refer to Section 5.3 (page 32) and Appendix B (page 62-70) of the Student Handbook)* for any kind of copying or collaboration on any assignment."

Liang Yrenn Da (13/7/2024)

"I *Howell Yek Teck Kee* verify that this paper contains entirely my own work. I have not consulted with any outside person or materials other than what was specified (an interviewee, for example) in the assignment or the syllabus requirements. Further, I have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. I realize the penalties *(refer to Section 5.3 (page 32) and Appendix B (page 62-70) of the Student Handbook)* for any kind of copying or collaboration on any assignment."

Howell Yek Teck Kee (13/7/2024)

"I Ooi Jin Yon verify that this paper contains entirely my own work. I have not consulted with any outside person or materials other than what was specified (an interviewee, for example) in the assignment or the syllabus requirements. Further, I have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. I realize the penalties *(refer to Section 5.3 (page 32) and Appendix B (page 62-70) of the Student Handbook)* for any kind of copying or collaboration on any assignment."

Ooi Jin Yon (13/7/2024)

"I *Yeoh Jie Long* verify that this paper contains entirely my own work. I have not consulted with any outside person or materials other than what was specified (an interviewee, for example) in the assignment or the syllabus requirements. Further, I have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. I realize the penalties *(refer to Section 5.3 (page 32) and Appendix B (page 62-70) of the Student Handbook)* for any kind of copying or collaboration on any assignment."

Yeoh Jie Long (13/7/2024)

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http://sunway.edu.my/pdpa/notice_bm (Malay version)

1.0 – Introduction

The dataset chosen for this assignment focuses on **health insurance price prediction**. This dataset records the insurance price value based on an individual's health status and condition. It contains **seven variables** with their respective category of data, including:

AGE	Age of the client (in years)
SEX	Gender of the client (male or female)
BMI	Body mass index (BMI) of the client
CHILDREN	The number of children that the client have
SMOKER	Smoking status of the client (yes or no)
REGION	The region where the client stays in the US
CHARGES	Health insurance charges purchased by the client

Besides that, the dataset we selected was relevant to **Sustainable Development Goal (SDG) 3: Good Health and Well-being**. SDG Goal 3 is to ensure everyone, regardless of age, leads a healthy life and experiences overall well-being (World Health Organization, 2024). SDG Goal 3 aims to ensure healthy lives and promote well-being, which is highly related to the public's insurance purchases. By analyzing health insurance charges, we can gain insights into how various factors can impact their health insurance costs. In this assignment, we will look into the relationship between health insurance charges and factors such as age, sex, BMI, number of students, smoking status, and living region. In this assessment, we successfully applied the techniques we learned in class, including one sample hypothesis, two sample hypotheses, ANOVA, ANCOVA,one-way ANOVA, two-way ANOVA and regression to understand the message brought out by the dataset through SAS Enterprise Guide.

2.0 - Descriptive Analysis

2.1 - Numerical Data

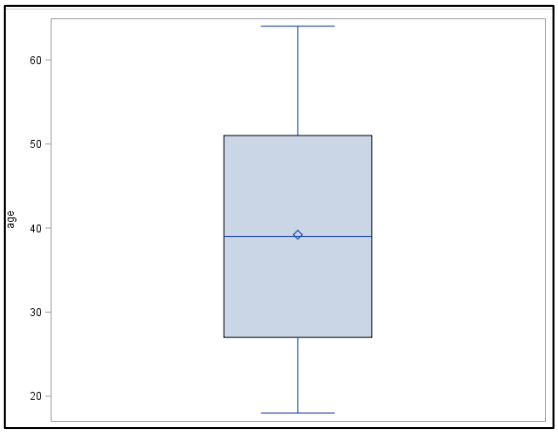
2.1.1 – Age

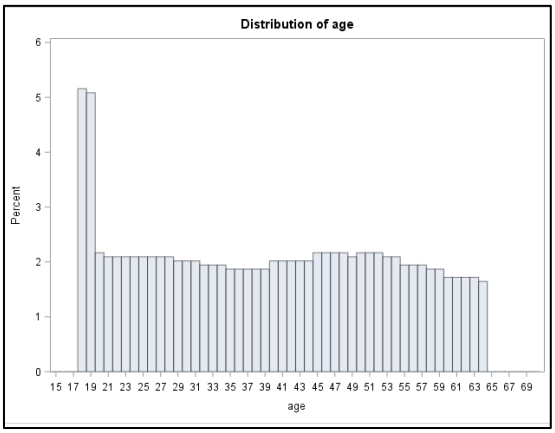
Summary of AGE attribute

Results

The MEANS Procedure

Analysis Variable : age									
Mean	Std Dev	Minimum	Maximum	Mode	Range	N	Lower Quartile	Median	Upper Quartile
39.2070254	14.0499604	18.0000000	64.0000000	18.0000000	46.0000000	1338	27.0000000	39.0000000	51.0000000





The dataset indicates that the public is very concerned about the importance of health insurance since they are small. It is because the dataset shows the mode of 18-year-olds that the public starts to purchase health insurance once they grow up. The number of clients who purchase health insurance between the ages of 20 and 64 is very consistent, which indicates that the public is conscious of paying for their health insurance to protect security when an accident arises.

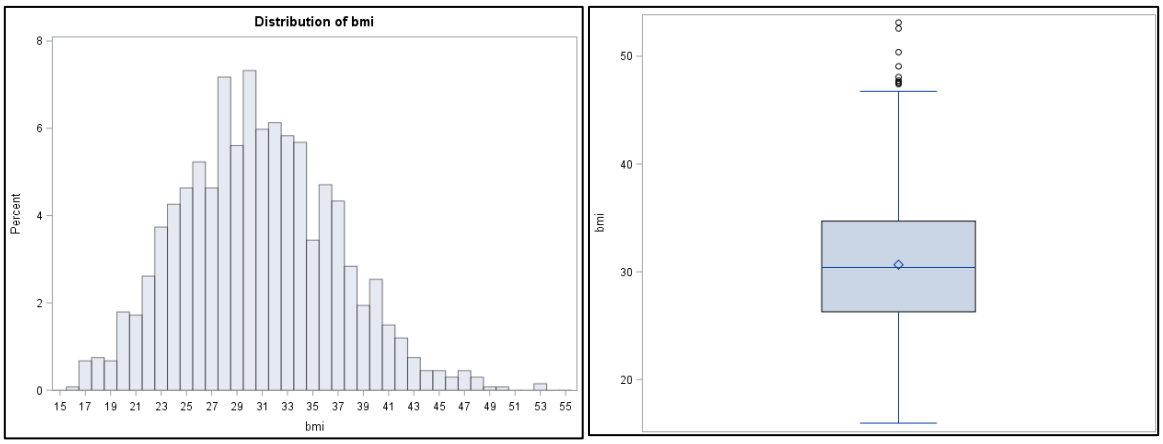
2.1.2 - BMI

Summary of BMI attribute

Results

The MEANS Procedure

Analysis Variable : bmi									
Mean	Std Dev	Minimum	Maximum	Mode	Range	N	Lower Quartile	Median	Upper Quartile
30.6633969	6.0981869	15.9600000	53.1300000	32.3000000	37.1700000	1338	26.2900000	30.4000000	34.7000000



The above table shows that the mean BMI of customers in the dataset is 30.66, slightly higher than the normal range. According to BetterHealth Channel, the normal range BMI should be between 18.5 and 24.9. From the histogram and box plot, we can indicate that it is slightly right-skewed and has outliers, which we predict suggest that the sample in the dataset may not have a balanced diet and may not exercise regularly.

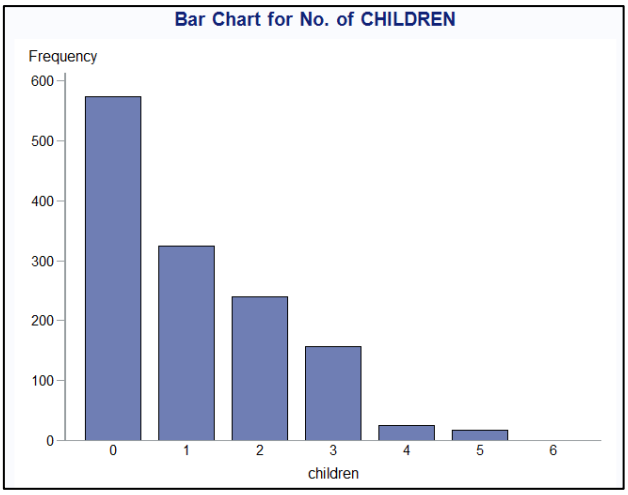
2.1.3 - Children

Summary of CHILDREN attribute

Results

The MEANS Procedure

Analysis Variable : children									
Mean	Std Dev	Minimum	Maximum	Mode	Range	N	Lower Quartile	Median	Upper Quartile
1.0949178	1.2054927	0	5.0000000	0	5.0000000	1338	0	1.0000000	2.0000000



The bar chart shows that most clients choose not to have their child. When the number of children increases, the frequency decreases. The bar charts prove that the frequency continuously decreases from 0, 1, 2, 3, 4 number of children and lastly, the customer with 5 number of children is the least.

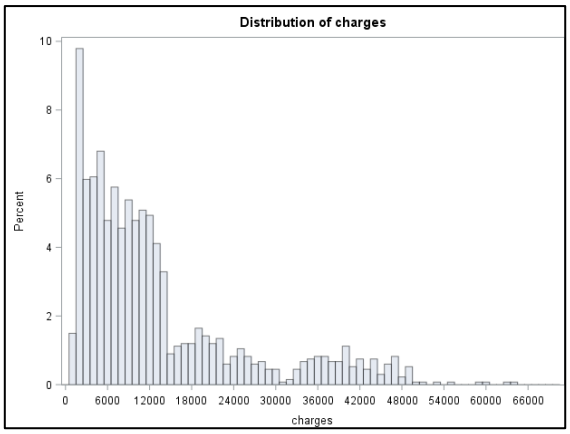
2.1.4 - Charges

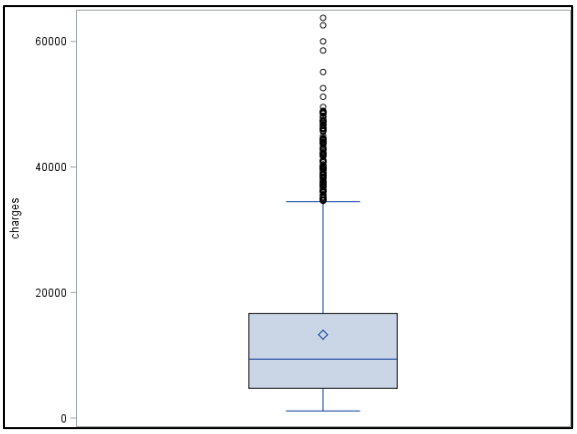
Summary of CHARGES attribute

Results

The MEANS Procedure

Analysis Variable : charges									
Mean	Std Dev	Minimum	Maximum	Mode	Range	N	Lower Quartile	Median	Upper Quartile
13270.42	12110.01	1121.87	63770.43	1639.56	62648.55	1338	4738.27	9382.03	16657.72





The above table shows that the mean charges of clients in the dataset are $13270.42. The range of health insurance purchased by the customers is $62648.55, with the lowest customer charges of $1121.87 and the highest of $63770.43. The data has an extensive spread with a standard deviation of $12110.01 because of the different economic statuses and purchase abilities.

The histogram above shows the charges for the health insurance the customer purchased, with a gap of $1000 in between. The histogram tells us that most customers purchased from $2000 to $2999. The histogram is right and positively skewed. With the calculations, the upper limit of the box and whisker plot is $34536.90, and 139 customers are out of this range.

CHARGES more than upper limit

Results

The MEANS Procedure

Analysis Variable : charges		
Minimum	Maximum	N
34617.84	63770.43	139

2.1.5 - Summary

Summary Statistics

Results

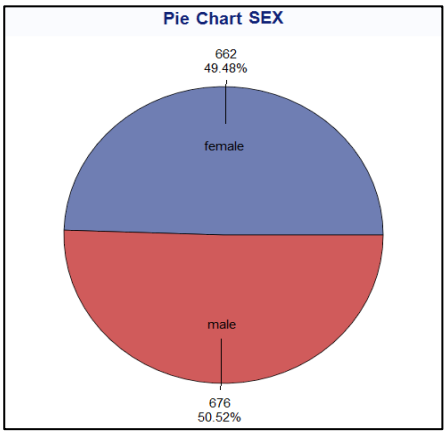
The MEANS Procedure

Variable	Mean	Std Dev	Minimum	Maximum	Range	N	N Miss	Lower Quartile	Median	Upper Quartile
age	39.2070254	14.0499604	18.0000000	64.0000000	46.0000000	1338	0	27.0000000	39.0000000	51.0000000
children	1.0949178	1.2054927	0	5.0000000	5.0000000	1338	0	0	1.0000000	2.0000000
bmi	30.6633969	6.0981869	15.9600000	53.1300000	37.1700000	1338	0	26.2900000	30.4000000	34.7000000
charges	13270.42	12110.01	1121.87	63770.43	62648.55	1338	0	4738.27	9382.03	16657.72

In summary, this dataset has 1338 data points and four numerical attributes: age, children, BMI, and charges. No missing value is included in this dataset, and all the outliers are natural outliers as the living conditions and purchase ability are different.

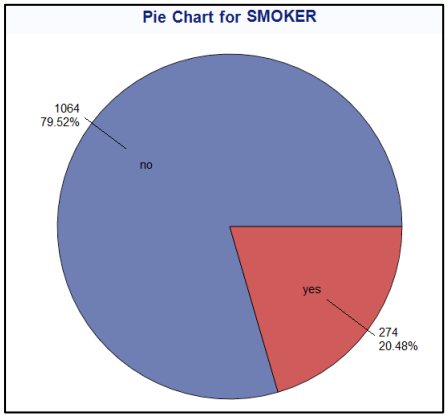
2.2 - Categorical Data

2.2.1 - Sex



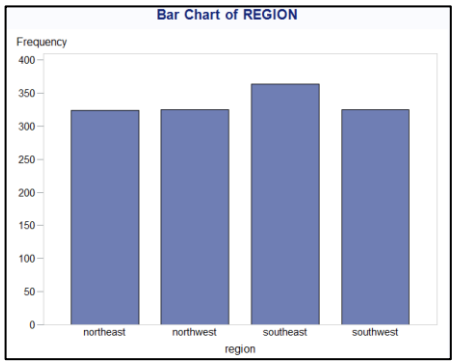
Sex can be classified into male and female. In this dataset, the ratio of males and females is well distributed and nearly the same: 50.52% and 49.48% respectively.

2.2.2 - Smoker



In this dataset, the ratio of smokers to non-smokers is approximately 4:1, with 79.52% smokers and 20.48% non-smokers, which shows that the majority knows that smoking will damage their health condition and avoid it.

2.2.3 - Region



The region is classified into four areas: northwest, northeast, southeast, and southwest. In this dataset, the distribution is relatively even, with the largest at 364 and the most minor at 324.

3.0 - Statistical Inference and Modelling

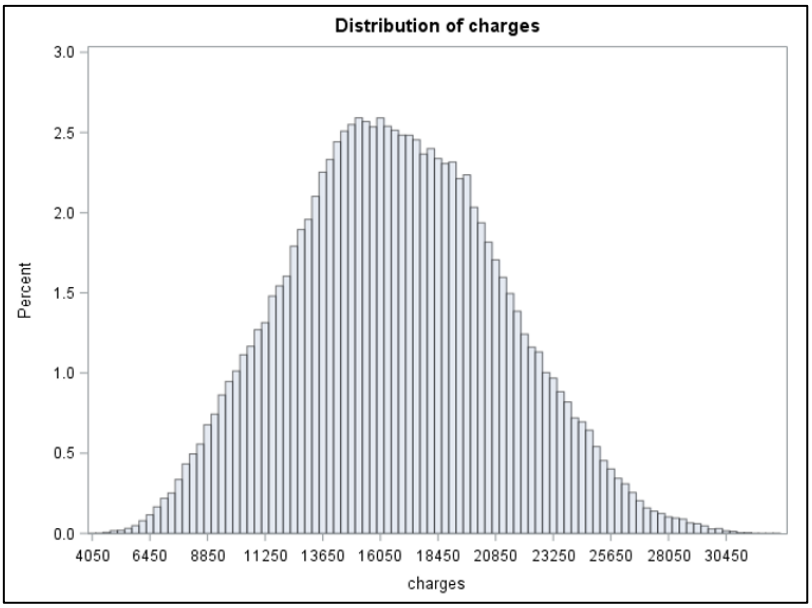
3.1 – One Sample Hypothesis

The health insurance charges of population in the US

SUMMARY STATISTICS - COMAPARE DATA

The MEANS Procedure

Analysis Variable : charges						
Mean	Std Dev	Minimum	Maximum	N	Lower 95% CL for Mean	Upper 95% CL for Mean
16726.94	4407.60	4056.64	32561.56	200000	16707.62	16746.25



The mean value of the US population's health insurance charges is $16726.94, and the standard deviation is $4407.60.

$$\mu = \$16726.94$$

$$\alpha = \$4407.60$$

Assume that we form our dataset from the above population through random selection. Does the data provide sufficient evidence to support that the mean health insurance charges of both the population and sample are similar? Use a significant level of $\alpha = 0.05$.

Health Insurance - T Test Result

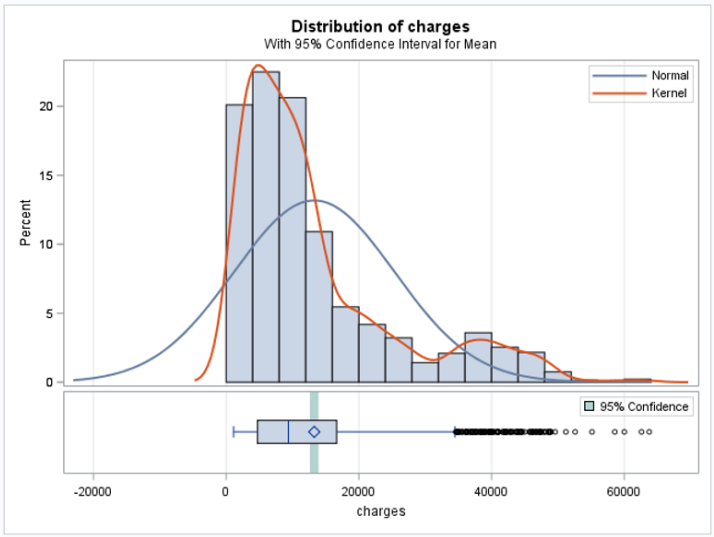
The TTEST Procedure

Variable: charges

N	Mean	Std Dev	Std Err	Minimum	Maximum
1338	13270.4	12110.0	331.1	1121.9	63770.4

Mean	95% CL Mean		Std Dev	95% CL Std Dev	
13270.4	12621.0	13919.9	12110.0	11667.9	12587.2

DF	t Value	Pr > |t|
1337	-10.44	<.0001



$$H_0: \mu = \$16726.94$$

$$H_1: \mu \neq \$16726.94$$

The test resulted in a t-value of -10.44 and a p-value of < 0.001. This indicates that we reject the null hypothesis and conclude that there is a significant difference between the mean of our data and the population mean health insurance charges in the US. There is sufficient evidence supporting the alternative hypothesis.

3.2 – Two Sample Hypothesis

Are there any differences between the health insurance charges of male and female clients? Use the significant level of $\alpha = 0.05$.

t Test

The TTEST Procedure

Variable: charges

sex	N	Mean	Std Dev	Std Err	Minimum	Maximum
female	662	12569.6	11128.7	432.5	1607.5	63770.4
male	676	13956.8	12971.0	498.9	1121.9	62592.9
Diff (1-2)		-1387.2	12094.6	661.3		

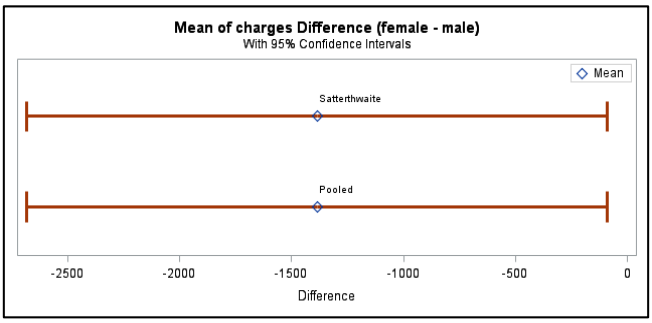
sex	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
female		12569.6	11720.3	13418.9	11128.7	10559.8	11762.9
male		13956.8	12977.2	14936.3	12971.0	12314.5	13702.0
Diff (1-2)	Pooled	-1387.2	-2684.5	-89.8123	12094.6	11653.0	12571.4
Diff (1-2)	Satterthwaite	-1387.2	-2682.5	-91.8553			

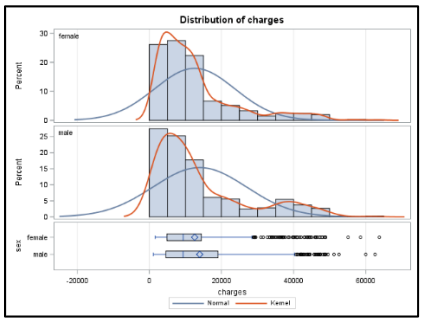
Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	1336	-2.10	0.0361
Satterthwaite	Unequal	1313.4	-2.10	0.0358

$$H_0: \mu_1 = \mu_2$$

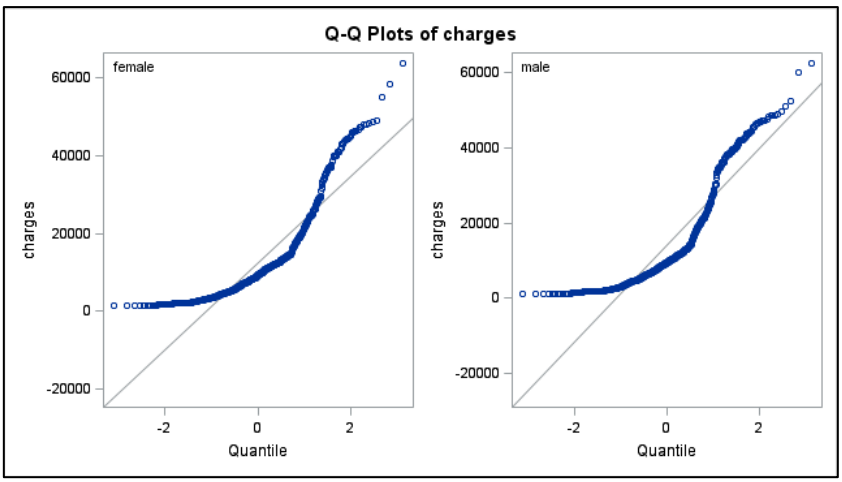
$$H_A: \mu_1 \neq \mu_2$$

As the testing equality of variances is rejected, we should use Satterthwaite (unpooled t-test) to solve the two-tail test. The p-value is 0.0358, less than the significant level of 0.05. Therefore, the null hypothesis is rejected. The mean insurance charges for males and females are not equal.





The graph visually confirms that the mean charges for males are significantly higher than for females.



The Q-Q plots show heavy tails and outliers, which indicate that the data have more extreme values than would be expected under a normal distribution.

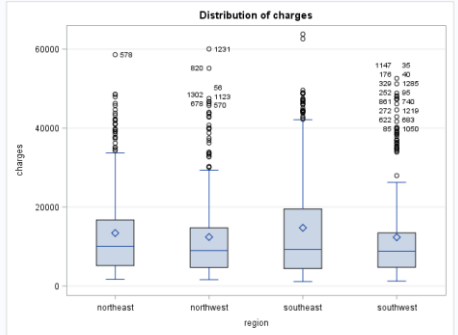
3.3 - ANOVA

3.3.1 - One-Way ANOVA

Is there any difference in the mean health insurance charges for the sample of different regions? Use a significant level of $\alpha = 0.05$.

Linear Models

The GLM Procedure



Level of region	N	charges Mean	charges Std Dev
northeast	324	13406.3845	11255.8031
northwest	325	12417.5754	11072.2769
southeast	364	14735.4114	13971.0986
southwest	325	12346.9374	11557.1791

There are only small differences between mean insurance charges from different regions. Both regions have a large spread of mean insurance charges, which causes many outliers based on their economic status and the type of insurance they buy.

$\boldsymbol{H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4}$ $\boldsymbol{(mean\ insurance\ charges\ of\ all\ regions\ are\ the\ same)}$

$\boldsymbol{H_1: At\ least\ two\ of\ the\ population\ means\ are\ different}$

Linear Models

The GLM Procedure

Class Level Information		
Class	Levels	Values
region	4	northeast northwest southeast southwest

Number of Observations Read	1338
Number of Observations Used	1338

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Page Break

Linear Models

The GLM Procedure

Dependent Variable: charges

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	1300759681.3	433586560.44	2.97	0.0309
Error	1334	194773461887	146007092.87		
Corrected Total	1337	196074221568			

R-Square	Coeff Var	Root MSE	charges Mean
0.006634	91.05467	12083.34	13270.42

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	3	1300759681	433586560	2.97	0.0309

The number of observations read and used is the same because there are no missing values for any variable in the model. The overall F-test was 2.97, and the p-value was 0.0309, lower than the significant level (0.05), so we rejected the null hypothesis. We concluded that there is a significant difference between the mean insurance charges of different regions. The coefficient of determination is 0.006634, which shows that Region only explains 0.66% of the variability of mean insurance charges.

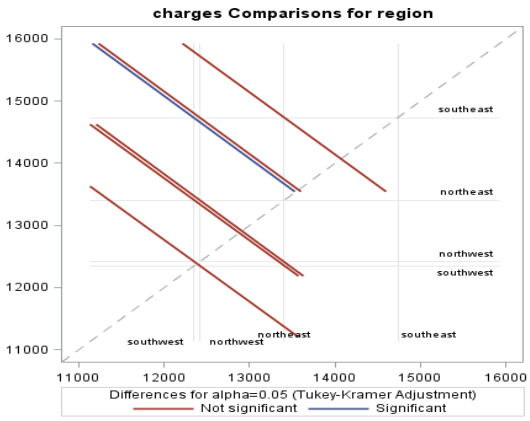
Linear Models

The GLM Procedure
Least Squares Means
Adjustment for Multiple Comparisons: Tukey-Kramer

region	charges LSMEAN	LSMEAN Number
northeast	13406.3845	1
northwest	12417.5754	2
southeast	14735.4114	3
southwest	12346.9374	4

Least Squares Means for effect region
Pr > |t| for H0: LSMean(i)=LSMean(j)
Dependent Variable: charges

i/j	1	2	3	4
1		0.7245	0.4745	0.6792
2	0.7245		0.0583	0.9999
3	0.4745	0.0583		0.0477
4	0.6792	0.9999	0.0477	



The region with the highest mean insurance charges is $14735.41, and the lowest is $12346.94. The only significant pairwise difference is between the southeast and southwest (p-value = 0.0477). The diffogram also shows the additional significant difference between the southeast and southwest.

```
PROC GLM DATA=WORK.SORTTempTableSorted
         PLOTS(ONLY)=ALL
    ;
    CLASS region;
    MODEL charges=  region
         /
         SS3
         SINGULAR=1E-07
    ;
    LSMEANS region / PDIFF=ALL ADJUST=TUKEY ;
    MEANS region /  HOVTEST=LEVENE (TYPE=SQUARE);
    RUN;
```

3.3.2 – Two-Way ANOVA

You are concerned that the sex of the client causes variability. Was adding blocking variable sex into the design detrimental to the test of the region?

Linear Models

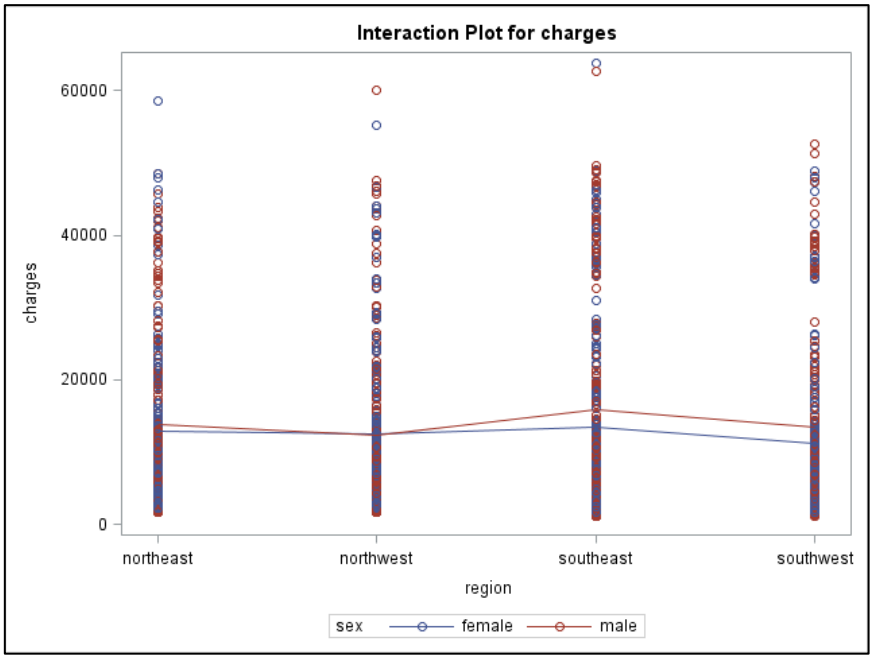
The GLM Procedure

Dependent Variable: charges

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	2254002194.1	322000313.44	2.21	0.0311
Error	1330	193820219374	145729488.25		
Corrected Total	1337	196074221568			

R-Square	Coeff Var	Root MSE	charges Mean
0.011496	90.96807	12071.85	13270.42

Source	DF	Type III SS	Mean Square	F Value	Pr > F
region	3	1253373965	417791322	2.87	0.0355
sex	1	584157098	584157098	4.01	0.0455
region*sex	3	340173346	113391115	0.78	0.5062



The overall significance of the model is indicated by the F Value (2.21) and probability 0.0311, which is less than the significant level of 0.05. H_0 is rejected. The model is statistically significant, where at least one of the predictors has a substantial effect on insurance charges.

R-Square = 0.0115

The model explains about 1.15% of the variable in charges, which is a very weak influence on explaining much of the variability in charges.

Root MSE = $12071.85

The typical difference between actual charges of insurance and the predicted insurance charges using the regression equation is approximately $12068.83

Region:

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$$H_1: At\ least\ one\ of\ the\ \mu_i\ is\ different, i = 1, \dots, 4$$

The p-value of 0.0355 is less than the significant level α = 0.05. H_0 is rejected. There is a significant difference between the regions.

Age:

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

The p-value of 0.0455 is less than the significant level α = 0.05. H_0 is rejected. There is a significant difference between the male and female.

Interaction of region and age:

H_0: **There is no interaction effect between sex and region**.

H_1: **There is an interaction effect between sex and region**.

The p-value for sex*region is 0.5062.

At α = 0.05, H_0 is not rejected. We conclude that there is no interaction between the two factors.

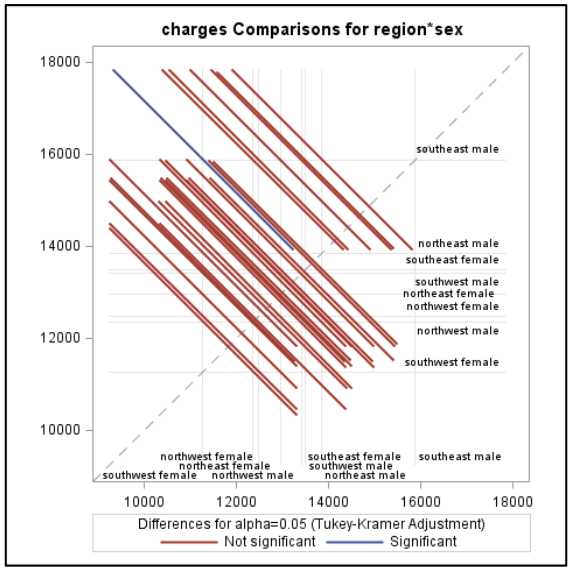
Linear Models

The GLM Procedure
Least Squares Means
Adjustment for Multiple Comparisons: Tukey-Kramer

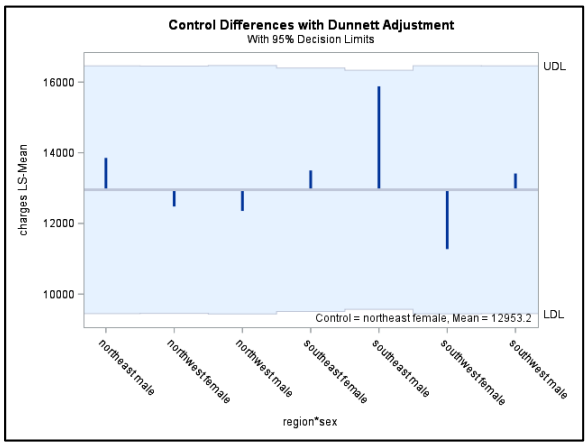
region	sex	charges LSMEAN	LSMEAN Number
northeast	female	12953.2032	1
northeast	male	13854.0054	2
northwest	female	12479.8704	3
northwest	male	12354.1196	4
southeast	female	13499.6692	5
southeast	male	15879.6172	6
southwest	female	11274.4113	7
southwest	male	13412.8836	8

Least Squares Means for effect region*sex
Pr > |t| for H0: LSMean(i)=LSMean(j)
Dependent Variable: charges

i/j	1	2	3	4	5	6	7	8
1		0.9977	1.0000	0.9998	0.9999	0.3166	0.9167	1.0000
2	0.9977		0.9700	0.9529	1.0000	0.7684	0.5330	1.0000
3	1.0000	0.9700		1.0000	0.9942	0.1430	0.9860	0.9970
4	0.9998	0.9529	1.0000		0.9887	0.1165	0.9929	0.9937
5	0.9999	1.0000	0.9942	0.9887		0.5655	0.6934	1.0000
6	0.3166	0.7684	0.1430	0.1165	0.5655		0.0090	0.5431
7	0.9167	0.5330	0.9860	0.9929	0.6934	0.0090		0.7524
8	1.0000	1.0000	0.9970	0.9937	1.0000	0.5431	0.7524	



The p-value for the comparison of southeast males and southeast females is 0.0090, less than $\alpha = 0.05$. There is a significant difference between them. The p-value for other comparisons is higher than $\alpha = 0.05$. There is no significant difference between them.



Least Squares Means
Adjustment for Multiple Comparisons: Dunnett

region	sex	charges LSMEAN	H0:LSMean=Control Pr > |t|
northeast	female	12953.2032	
northeast	male	13854.0054	0.9776
northwest	female	12479.8704	0.9995
northwest	male	12354.1196	0.9980
southeast	female	13499.6692	0.9987
southeast	male	15879.6172	0.1201
southwest	female	11274.4113	0.6858
southwest	male	13412.8836	0.9996

The Dunnett adjustment diagram and the probability show no significant difference, as no line extends beyond the shaded area.

```
PROC GLM DATA=WORK.SORTTempTableSorted
        PLOTS(ONLY)=ALL
    ;
    CLASS region sex;
    MODEL charges=  region sex region*sex
        /
        SS3
        SINGULAR=1E-07
    ;
    LSMEANS region sex region*sex / PDIFF=ALL ADJUST=TUKEY ;
    LSMEANS region sex region*sex / PDIFF=ALL ADJUST=T ;
    LSMEANS region sex region*sex / PDIFF=CONTROL ;
    MEANS region sex region*sex / TUKEY  HOVTEST=LEVENE (TYPE=SQUARE);
    RUN;
```

3.4 Linear Regression

Linear Models

The GLM Procedure

Dependent Variable: charges

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	147234688724	18404336091	500.81	<.0001
Error	1329	48839532844	36749084.156		
Corrected Total	1337	196074221568			

R-Square	Coeff Var	Root MSE	charges Mean
0.750913	45.68131	6062.102	13270.42

Source	DF	Type III SS	Mean Square	F Value	Pr > F
children	1	437546614.34	437546614.34	11.91	0.0006
bmi	1	5169225279.4	5169225279.4	140.66	<.0001
age	1	17124473621	17124473621	465.98	<.0001
region	3	233431209.22	77810403.074	2.12	0.0962
smoker	1	122446774088	122446774088	3331.97	<.0001
sex	1	5716429.0666	5716429.0666	0.16	0.6933

Parameter		Estimate		Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept		10818.63062	B	1071.814768	10.09	<.0001	8715.99736	12921.26387
children		475.50055		137.804093	3.45	0.0006	205.16329	745.83780
bmi		339.19345		28.599470	11.86	<.0001	283.08843	395.29848
age		256.85635		11.898849	21.59	<.0001	233.51378	280.19893
region	northeast	960.05099	B	477.933024	2.01	0.0448	22.46560	1897.63638
region	northwest	607.08709	B	477.203912	1.27	0.2035	-329.06796	1543.24215
region	southeast	-74.97106	B	470.638641	-0.16	0.8735	-998.24669	848.30457
region	southwest	0.00000	B	.	.	.	.	.
smoker	no	-23848.53454	B	413.153355	-57.72	<.0001	-24659.03838	-23038.03071
smoker	yes	0.00000	B	.	.	.	.	.
sex	female	131.31436	B	332.945439	0.39	0.6933	-521.84155	784.47027
sex	male	0.00000	B	.	.	.	.	.

H_0: β (charges) = β (children) = β (bmi) = β(age) = β (region northeast) = β (region northwest) = β (region southeast) = β (smoker no) = β (sex female) = 0

H_1: At least one β ≠ 0.

The F-value is 500.81, corresponding p-value <0.0001.

The p-value is < 0.0001, which is smaller than the significance value (α = 0.05).

Hence, H_0 is rejected.

At least one of the explanatory variables has a significant effect on the response variable, charges.

The standard error of the estimate is reported by Root MSE = 6062.102

The typical difference between actual charges and the predicted charges using the regression equation is approximately $6062.102.

The regression equation for these data is:

$$\hat{Y} = 10818.63062 + 475.50055\,x_1 + 339.19345x_2 + 256.85635x_3 + 960.05099x_4 + 607.08709x_5 - 74.97106x_6 - 23848.53454x_7 + 131.31436x_8$$

The equation describes the effect of an independent variable controlling the effects of the other independent variables in the model:

$\hat{Y}$: charges

X1: children

X2: BMI

X3: age

X4: region (northeast)

X5: region (northwest)

X6: region (southeast)

X7: smoker (no)

X8: sex (female)

475.50055 is interpreted as the change in charges (Y) corresponding to the change in children (X1) when BMI (X2), age (X3), region northeast (X4) and region northwest (X5), region southeast (X6), smoker no (X7), and sex female (X8) are held constant.

When the BMI (X2) increases by one unit, the charges (Y) increase by 339.19345, when children (X1), age (X3), region northeast (X4) and region northwest (X5), region southeast (X6), smoker no (X7), and sex female (X8) are held constant.

When the age (X3) increases by one year, the charges (Y) increase by 256.85635, when children (X1), BMI (X2), region northeast (X4) and region northwest (X5), region southeast (X6), smoker no (X7), and sex female (X8) are held constant.

When the region northeast (X4) increases, the charges (Y) increase by 960.05099, when children (X1), BMI (X2), age (X3), region northwest (X5), region southeast (X6), smoker no (X7), sex female (X8) are held constant.

When the region northwest (X5) increases, the charges (Y) increase by 607.08709, when children (X1), bmi (X2), age (X3), region northeast (X4), region southeast (X6), smoker no (X7), sex female (X8) are held constant.

When the region sorthwest (X6) increases by one unit, the charges (Y) decrease by 74.97106, when children (X1), bmi (X2), age (X3), region northeast (X4), region northwest (X5), smoker no (X7), sex female (X8) are held constant.

When the smoker no (X7) is true(1), the charges (Y) decrease by 23848.53454, when children (X1), bmi (X2), age (X3), region northeast (X4), region northwest (X5), region southwest (X6), sex female (X8) are held constant.

When the sex female (X8) increases, the charges (Y) increase by 131.31436, when children (X1), bmi (X2), age (X3), region northeast (X4), region northwest (X5), region southwest (X6), smoker no (X7) are held constant.

3.4.2 Children

Parameter		Estimate		Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept		10818.63062	B	1071.814768	10.09	<.0001	8715.99736	12921.26387
children		475.50055		137.804093	3.45	0.0006	205.16329	745.83780
bmi		339.19345		28.599470	11.86	<.0001	283.08843	395.29848
age		256.85635		11.898849	21.59	<.0001	233.51378	280.19893
region	northeast	960.05099	B	477.933024	2.01	0.0448	22.46560	1897.63638
region	northwest	607.08709	B	477.203912	1.27	0.2035	-329.06796	1543.24215
region	southeast	-74.97106	B	470.638641	-0.16	0.8735	-998.24669	848.30457
region	southwest	0.00000	B	.	.	.	.	.
smoker	no	-23848.53454	B	413.153355	-57.72	<.0001	-24659.03838	-23038.03071
smoker	yes	0.00000	B	.	.	.	.	.
sex	female	131.31436	B	332.945439	0.39	0.6933	-521.84155	784.47027
sex	male	0.00000	B	.	.	.	.	.

H_0: β(children) = 0

H_1: β(children) ≠ 0

The test statistics t-value = 3.45, the corresponding p-value is 0.0006,

H_0 is rejected at significance level α = 0.05.

There is strong evidence that children are related to the charges, controlling for BMI, age, region northeast, region northwest, region southeast, smoker no, and sex female.

Controlling for other independent variables in the model, we are 95% confident that the change in mean charges increases in children falls between 205.16329 and 745.83780.

3.4.3 BMI

Parameter		Estimate		Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept		10818.63062	B	1071.814768	10.09	<.0001	8715.99736	12921.26387
children		475.50055		137.804093	3.45	0.0006	205.16329	745.83780
bmi		339.19345		28.599470	11.86	<.0001	283.08843	395.29848
age		256.85635		11.898849	21.59	<.0001	233.51378	280.19893
region	northeast	960.05099	B	477.933024	2.01	0.0448	22.46560	1897.63638
region	northwest	607.08709	B	477.203912	1.27	0.2035	-329.06796	1543.24215
region	southeast	-74.97106	B	470.638641	-0.16	0.8735	-998.24669	848.30457
region	southwest	0.00000	B	.	.	.	.	.
smoker	no	-23848.53454	B	413.153355	-57.72	<.0001	-24659.03838	-23038.03071
smoker	yes	0.00000	B	.	.	.	.	.
sex	female	131.31436	B	332.945439	0.39	0.6933	-521.84155	784.47027
sex	male	0.00000	B	.	.	.	.	.

H_0: β(bmi) = 0

H_1: β(bmi) $\neq$ 0

The test statistics t-value = 11.86 with corresponding p-value is <0.0001,

H_0 is rejected at significance level α = 0.05.

There is strong evidence that BMI is related to the charges, controlling for children, age, region northeast, region northwest, region southeast, smoker no, and sex female.

Controlling for other independent variables in the model, we are 95% confident that the change in mean charges increases in BMI falls between 283.08843 and 395.29848.

3.4.4 Age

Parameter		Estimate		Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept		10818.63062	B	1071.814768	10.09	<.0001	8715.99736	12921.26387
children		475.50055		137.804093	3.45	0.0006	205.16329	745.83780
bmi		339.19345		28.599470	11.86	<.0001	283.08843	395.29848
age		256.85635		11.898849	21.59	<.0001	233.51378	280.19893
region	northeast	960.05099	B	477.933024	2.01	0.0448	22.46560	1897.63638
region	northwest	607.08709	B	477.203912	1.27	0.2035	-329.06796	1543.24215
region	southeast	-74.97106	B	470.638641	-0.16	0.8735	-998.24669	848.30457
region	southwest	0.00000	B	.	.	.	.	.
smoker	no	-23848.53454	B	413.153355	-57.72	<.0001	-24659.03838	-23038.03071
smoker	yes	0.00000	B	.	.	.	.	.
sex	female	131.31436	B	332.945439	0.39	0.6933	-521.84155	784.47027
sex	male	0.00000	B	.	.	.	.	.

H_0: β(age) = 0

H_1: β(age) ≠ 0

The test statistics t-value = 21.59 with corresponding p-value is <0.0001,

H_0 is rejected at significance level α = 0.05.

There is strong evidence that age is related to the charges, controlling for children, BMI, region northeast, region northwest, region southeast, smoker no, and sex female.

Controlling for other independent variables in the model, we are 95% confident that the change in mean charges increase in age falls between 233.51378 and 280.19893.

3.4.5 Region (northeast)

Parameter		Estimate		Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept		10818.63062	B	1071.814768	10.09	<.0001	8715.99736	12921.26387
children		475.50055		137.804093	3.45	0.0006	205.16329	745.83780
bmi		339.19345		28.599470	11.86	<.0001	283.08843	395.29848
age		256.85635		11.898849	21.59	<.0001	233.51378	280.19893
region	northeast	960.05099	B	477.933024	2.01	0.0448	22.46560	1897.63638
region	northwest	607.08709	B	477.203912	1.27	0.2035	-329.06796	1543.24215
region	southeast	-74.97106	B	470.638641	-0.16	0.8735	-998.24669	848.30457
region	southwest	0.00000	B	.	.	.	.	.
smoker	no	-23848.53454	B	413.153355	-57.72	<.0001	-24659.03838	-23038.03071
smoker	yes	0.00000	B	.	.	.	.	.
sex	female	131.31436	B	332.945439	0.39	0.6933	-521.84155	784.47027
sex	male	0.00000	B	.	.	.	.	.

H_0: β(region northeast) = 0

H_1: β(region northeast) $\neq$ 0

The test statistics t-value = 2.01 with corresponding p-value is 0.0448,

H_0 is rejected at significance level α = 0.05.

There is moderate evidence that region northeast is related to the charges, controlling for children, BMI, age, region northwest, region southeast, smoker no, sex female.

Controlling for other independent variables in the model, we are 95% confident that the change in mean charges increases in a northeast region falls between 22.46560 and 1897.63638.

3.4.6 Region (Northwest)

Parameter		Estimate		Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept		10818.63062	B	1071.814768	10.09	<.0001	8715.99736	12921.26387
children		475.50055		137.804093	3.45	0.0006	205.16329	745.83780
bmi		339.19345		28.599470	11.86	<.0001	283.08843	395.29848
age		256.85635		11.898849	21.59	<.0001	233.51378	280.19893
region	northeast	960.05099	B	477.933024	2.01	0.0448	22.46560	1897.63638
region	northwest	607.08709	B	477.203912	1.27	0.2035	-329.06796	1543.24215
region	southeast	-74.97106	B	470.638641	-0.16	0.8735	-998.24669	848.30457
region	southwest	0.00000	B	.	.	.	.	.
smoker	no	-23848.53454	B	413.153355	-57.72	<.0001	-24659.03838	-23038.03071
smoker	yes	0.00000	B	.	.	.	.	.
sex	female	131.31436	B	332.945439	0.39	0.6933	-521.84155	784.47027
sex	male	0.00000	B	.	.	.	.	.

H_0: β(region northwest) = 0

H_1: β(region northwest) $\neq$ 0

The test statistics t-value = 1.27 with corresponding p-value is 0.2035,

H_0 is not rejected at significance level α = 0.05.

There is weak evidence that region northwest is related to the charges, controlling for children, bmi, age, region northeast, region southeast, smoker no, sex female.

Controlling for other independent variables in the model, we are 95% confident that the change in mean charges increases in a northeast region falls between -329.06796 and 1543.24215.

3.4.7 Region (southeast)

Parameter		Estimate		Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept		10818.63062	B	1071.814768	10.09	<.0001	8715.99736	12921.26387
children		475.50055		137.804093	3.45	0.0006	205.16329	745.83780
bmi		339.19345		28.599470	11.86	<.0001	283.08843	395.29848
age		256.85635		11.898849	21.59	<.0001	233.51378	280.19893
region	northeast	960.05099	B	477.933024	2.01	0.0448	22.46560	1897.63638
region	northwest	607.08709	B	477.203912	1.27	0.2035	-329.06796	1543.24215
region	southeast	-74.97106	B	470.638641	-0.16	0.8735	-998.24669	848.30457
region	southwest	0.00000	B	.	.	.	.	.
smoker	no	-23848.53454	B	413.153355	-57.72	<.0001	-24659.03838	-23038.03071
smoker	yes	0.00000	B	.	.	.	.	.
sex	female	131.31436	B	332.945439	0.39	0.6933	-521.84155	784.47027
sex	male	0.00000	B	.	.	.	.	.

H_0: β(region southeast) = 0

H_1: β(region southeast) $\neq$ 0

The test statistics t-value = -0.16 with corresponding p-value is 0.8735,

H_0 is not rejected at significance level α = 0.05.

There is weak evidence that region southeast is related to the charges, controlling for children, BMI, age, region northeast, region northwest, smoker no, sex female.

Controlling for other independent variables in the model, we are 95% confident that the change in mean charges increases in the southeast region falls between -998.24669 and 848.30457.

3.4.8 Smoker (no)

Parameter		Estimate		Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept		10818.63062	B	1071.814768	10.09	<.0001	8715.99736	12921.26387
children		475.50055		137.804093	3.45	0.0006	205.16329	745.83780
bmi		339.19345		28.599470	11.86	<.0001	283.08843	395.29848
age		256.85635		11.898849	21.59	<.0001	233.51378	280.19893
region	northeast	960.05099	B	477.933024	2.01	0.0448	22.46560	1897.63638
region	northwest	607.08709	B	477.203912	1.27	0.2035	-329.06796	1543.24215
region	southeast	-74.97106	B	470.638641	-0.16	0.8735	-998.24669	848.30457
region	southwest	0.00000	B	.	.	.	.	.
smoker	no	-23848.53454	B	413.153355	-57.72	<.0001	-24659.03838	-23038.03071
smoker	yes	0.00000	B	.	.	.	.	.
sex	female	131.31436	B	332.945439	0.39	0.6933	-521.84155	784.47027
sex	male	0.00000	B	.	.	.	.	.

H_0: β(smoker no) = 0

H_1: β(smoker no) ≠ 0

The test statistics t-value = -57.72 with corresponding p-value is < 0.0001,

H_0 is rejected at significance level α = 0.05.

There is strong evidence that smoker no is related to the charges, controlling for children, bmi, age, region northeast, region northwest, region southeast, sex female.

Controlling for other independent variables in the model, we are 95% confident that the change in mean charges increase in southeast region falls between -24659.03838 and -23038.03071.

3.4.9 Sex (female)

Parameter		Estimate		Standard Error	t Value	Pr > |t|	95% Confidence Limits	
Intercept		10818.63062	B	1071.814768	10.09	<.0001	8715.99736	12921.26387
children		475.50055		137.804093	3.45	0.0006	205.16329	745.83780
bmi		339.19345		28.599470	11.86	<.0001	283.08843	395.29848
age		256.85635		11.898849	21.59	<.0001	233.51378	280.19893
region	northeast	960.05099	B	477.933024	2.01	0.0448	22.46560	1897.63638
region	northwest	607.08709	B	477.203912	1.27	0.2035	-329.06796	1543.24215
region	southeast	-74.97106	B	470.638641	-0.16	0.8735	-998.24669	848.30457
region	southwest	0.00000	B	.	.	.	.	.
smoker	no	-23848.53454	B	413.153355	-57.72	<.0001	-24659.03838	-23038.03071
smoker	yes	0.00000	B	.	.	.	.	.
sex	female	131.31436	B	332.945439	0.39	0.6933	-521.84155	784.47027
sex	male	0.00000	B	.	.	.	.	.

H_0: β(sex female) = 0

H_1: β(sex female) $\neq$ 0

The test statistics t-value = 0.39 with corresponding p-value is 0.6933,

H_0 is not rejected at significance level α = 0.05.

There is weak evidence that the female sex is related to the charges, controlling for children, BMI, age, region northeast, region northwest, region southeast, smoker no.

Controlling for other independent variables in the model, we are 95% confident that the change in mean charges increase in sex falls between -521.84155 and 784.47027.

4.0 – Conclusion

In conclusion, we conducted hypothesis testing, ANOVA, and linear regression to identify significant factors influencing health insurance charges. The analyses revealed significant differences based on sex and region. The two-sample hypothesis test showed that mean charges for males are significantly higher than for females (p-value = 0.0358). The one-way ANOVA indicated significant regional differences (p-value = 0.0309), though region explains only 0.66% of the variability in charges. The two-way ANOVA confirmed significant differences between regions (p-value = 0.0355) and between sexes (p-value = 0.0455), with no interaction effect. The typical difference between actual and predicted charges is approximately $12,068.83. The linear regression model identified that the number of children, BMI, age, region, smoking status, and sex significantly influence charges. Charges increase with higher BMI and age, while not smoking leads to lower charges. The typical difference between actual and predicted charges is about $6062, indicating the model's accuracy.

5.0 - References

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